TEAM PROMAX’s PITCH DECK

WATTWISE: SOLAR SPLITS THAT FIT

A FORECAST-LED LOCAL SOLAR ENERGY ALLOCATION ENGINE

SIME DARBY HACKATHON: TOWNSHIP OF TOMORROW

TEAM PROMAX:

01 TEAM LEAD

SHERI ANNE CHONG TING TING · LEAD, FULL-STACK
UNIVERSITY OF SOUTHAMPTON MALAYSIA (UOSM)
satc1g24@soton.ac.uk
019-807 2503

02 MEMBERS

ONG HUI MIN · UI/UX DEVELOPER, RESEARCH
UOSM · hmo1e25@soton.ac.uk

JESS LAU XI EN · UI/UX DESIGNER, DEVELOPER
UOSM · xel1n24@soton.ac.uk

LIM POH SHIU · DATA ANALYST, QA
UOSM ·psl1e25@soton.ac.uk

1) PROBLEM STATEMENT

CREAM makes SDP pre-allocate each buyer's share of solar four months ahead.

120

days

Each buyer's share is filed in writing before the month starts, before its real demand is known.

9

sen/kWh

Access charge still paid by SDP to release solar a buyer didn’t use.

6%

of solar given away

Median waste of 6% under a fixed yearly split. One buyer pair in ten wastes 16% or more.

CREAM Guidelines 9.9, 9.10(e), 13.2 (Energy Commission, 2025; Skrine, 2025). Replay on real meters, 407 buyer pairs (Miller et al., 2020; WattWise, 2026).

Check these numbers: Appendix A (solar given away), re-runnable in the testing suite.

2) TARGET USER(S)

Who has the problem today, and what WattWise changes

01

CREAM aggregators

TODAY

SDP files every buyer's share 120 days ahead

WITH WATTWISE

Each split is sized to the buyer's predicted use.

02

Elmina homeowners

TODAY

Fixed splits waste solar, so SDP has limited room to add more roofs.

WITH WATTWISE

Up to 20% more homes can lease their roofs at no extra waste (modelled).

03

CREAM buyers

TODAY

Innovation Park today; a school, petrol stations and commercial hubs have shown interest

WITH WATTWISE

Each share comes from the buyer's own meters.

Buyer interest: The Malaysian Reserve (2025). Waste figures: median of 407 real buyer pairs (WattWise, 2026).

Check these numbers: Appendix A (solar given away) and Appendix C (room for 20% more roofs).

3) OUR SOLUTION: WATTWISE

A software layer that replaces a fixed yearly guess with a monthly plan.

ESTIMATE

Each buyer's use, from its own meter history

ALLOCATE

Share sized to the estimate; a slice held back for a fallback buyer, NEDA (New Enhanced Dispatch Arrangement)

FILE

Stamped with its 120-day notice date

LEARN

Each month checked against real use

THE RULE ANYONE CAN CHECK BY HAND

estimate = same month last year × (1 + trend)

No weather, no machine learning. Replay and rules: 02 Scheduler Module/replay.py (WattWise, 2026).

4) INNOVATION & DIFFERENTIATION

What gives WattWise a new approach?

	Makes & stores solar	Manages one building	Signs up roofs	Plans CREAM shares
Solarvest · GSPARX · Plus Xnergy	✓	–	–	–
Schneider Electric · Tanand	–	✓	–	–
Bright Energy · GSPARX	–	–	✓	–
WATTWISE				✓

WattWise works with these players inside the CREAM ecosystem, not against them.

Company roles summarised from their public descriptions; not independently verified.

5) HOW IT WORKS

Theme: Planet · Smart energy efficiency

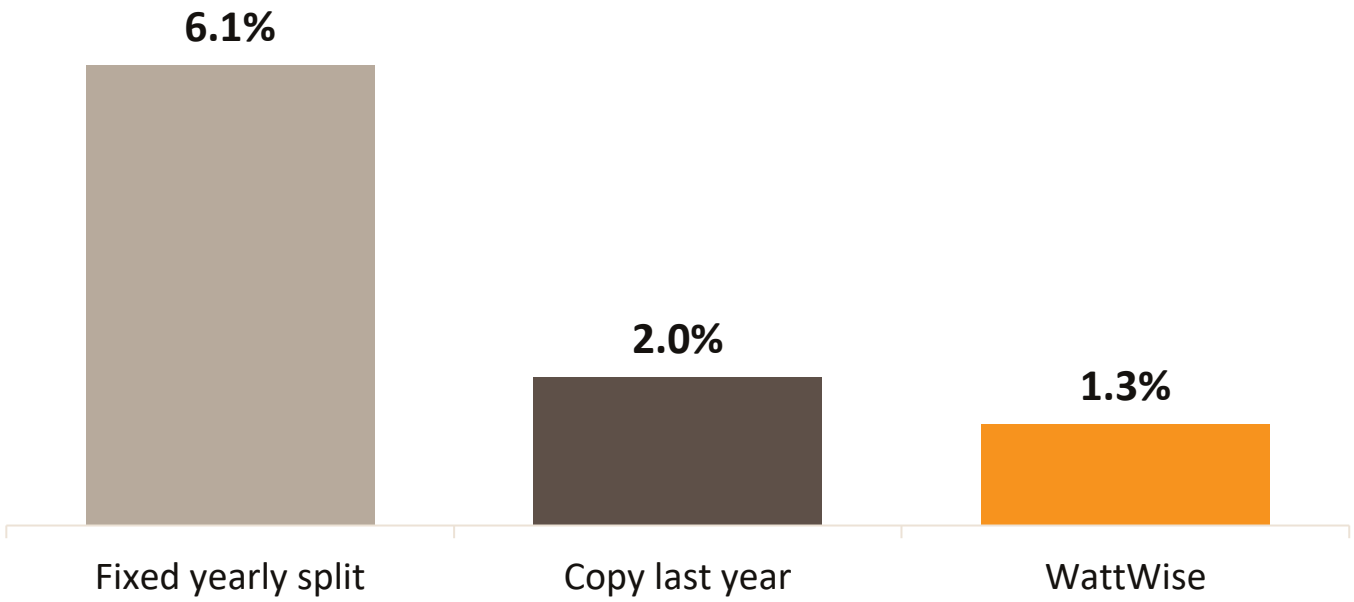
1	2	3	4	5
READ	**ESTIMATE**	**ALLOCATE**	**FILE**	**LEARN**
3 files in	**120 days ahead**	**share + hold-back**	**notice-stamped**	**after TNB bills**
meter readings · past splits · solar output	same month last year × trend	2% of solar held back for NEDA	SDP files with the Energy Commission	suggested vs filed vs actual

Result: solar given away drops from 6.05% to 1.34% (median, 407 buyer pairs).

Check these numbers: Appendix A (6.05% to 1.34% given away).

6) IMPACT

Less solar is thrown away



Median share of the year's solar given away from 407 real buyer pairs

78%

less solar given away than a fixed split

99.8%

of 407 real buyer pairs: WattWise gives away less than a fixed split

RM108k–265k

a year more at 6,000 homes. Scales with solar generated, not buyers (modelled)

For the 1,368-roof Elmina example (2 assumed buyers): 443 MWh a year kept local, about 330 tonnes of CO_2e avoided, and room for about 20% more roofs (modelled).

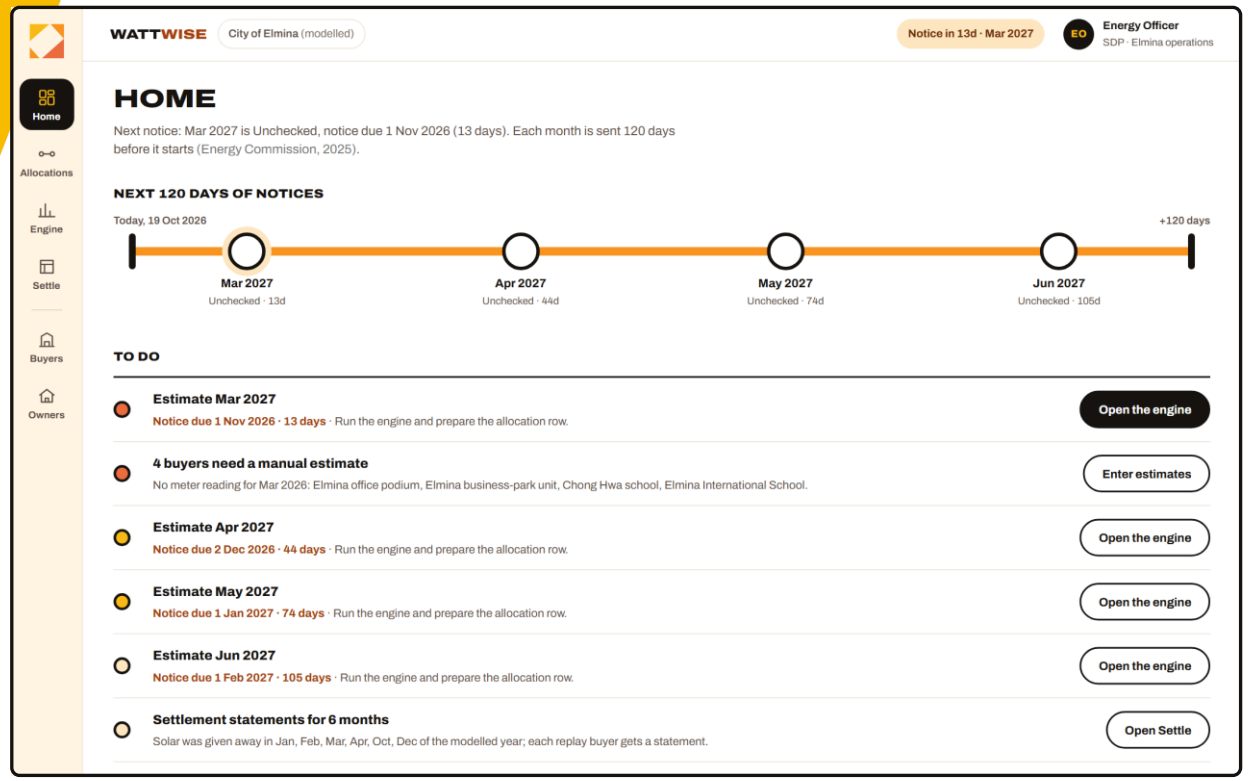
CO_2e at 0.740 t/MWh, Peninsular Malaysia grid, 2024 provisional (Energy Commission, 2026).

Real meters from 821 buildings (Miller et al., 2020) with rooftop solar at Elmina (European Commission JRC, 2022). Re-run it yourself in the testing suite.

Check these numbers: Appendix A (waste and CO_2e), B (income and how it scales), D (how many buyer pairs improve).

7) FEASIBILITY

A working prototype, tested on real meters



Try the testing suite: re-run every number

821 real buildings tested

99.8% of buyer pairs give away less than a fixed split

Resources: public data and a static web app; no hardware, no grid integration

Stakeholders: SDP scheme admin, Energy Commission, TNB (Single Buyer), CREAM buyers

Deployment: 3-month shadow pilot; SDP still files with the Energy Commission

Operator app prototype (screenshot). Fits SDP's digital energy services strategy (Sime Darby Property, 2026a).

Check these numbers: Appendix A (821 buildings) and Appendix D (99.8% of buyer pairs).

7) IMPLEMENTATION

How WattWise enters SDP's pipeline

01

Shadow pilot

3 months, free. SDP shares 3 files: meters, past splits, solar.

02

Verify

Compare suggested, filed and billed splits; report solar given away.

03

Live at Elmina

WattWise suggests, the scheme admin approves, SDP files 120 days ahead.

04

Scale

Run monthly for each new buyer; same rules, no new hardware.

05

Expand

Same rules in any CREAM township; more solar means more income.

Build to pilot: ~4–6 weeks

Estimate. Prototype works; left: SDP file import, notice export, security review

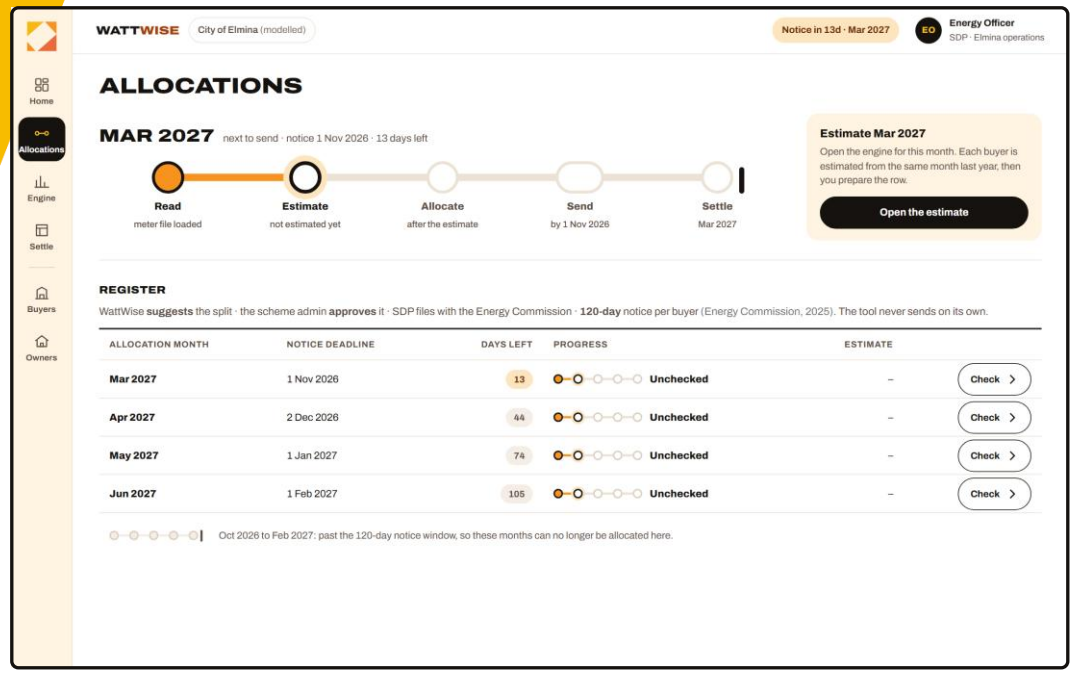
Upkeep: near zero

No hardware, no model to retrain; only update rates if the Energy Commission changes them

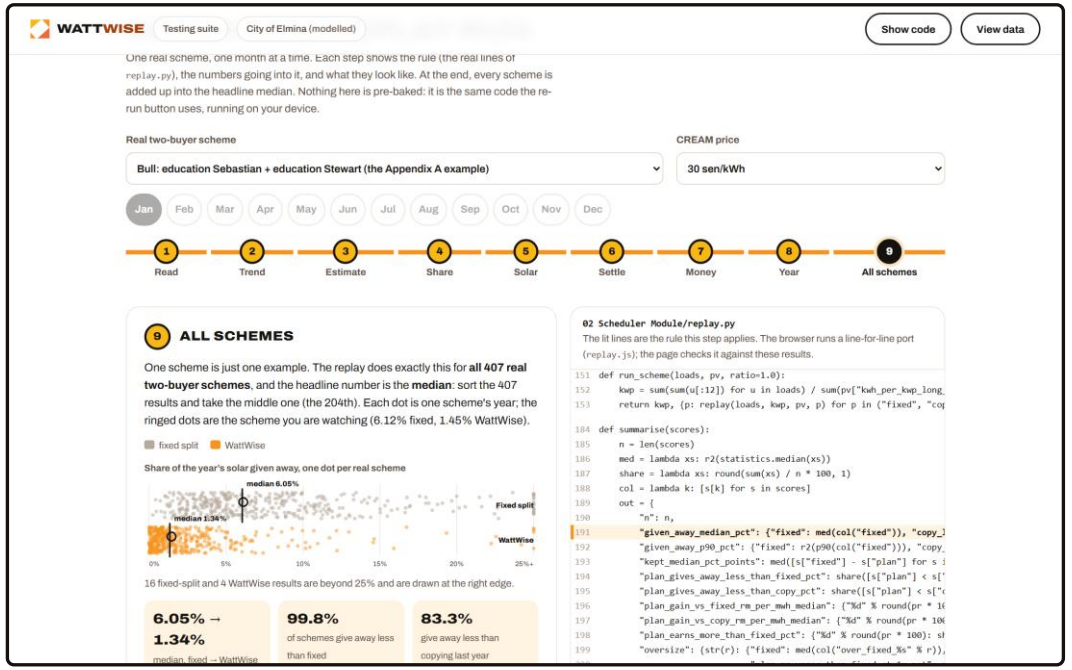
The first shadow report is back within a day (planned).

7) VISUAL CONCEPT

Explore the working prototype



Operator app: deadlines, estimate, notice



Testing suite: every claim, re-runnable

OPEN THE LIVE DASHBOARD

OPEN THE TESTING SUITE

Code and data: github.com/ttc-099/Wattwise

8) WHY WE SHOULD WIN

Malaysia’s grid is under growing pressure from data-centre demand and rising rooftop solar (Selan, 2026). CREAM lets residents’ solar serve nearby businesses, but fixed shares filed 120 days ahead send about 6% of it back to the grid unused. WattWise sizes each share from the buyer’s own meter history and cuts that to about 1.3%, so more local solar is used locally. It deserves recognition because it’s practical and checkable: a working prototype, tested on real meters from 821 buildings, with every number re-runnable by judges.



Sheri Anne Chong Ting Ting

Lead, full-stack



Ong Hui Min

UI/UX developer, research



Jess Lau Xi En

UI/UX designer, developer



Lim Poh Shiu

Data analyst, QA

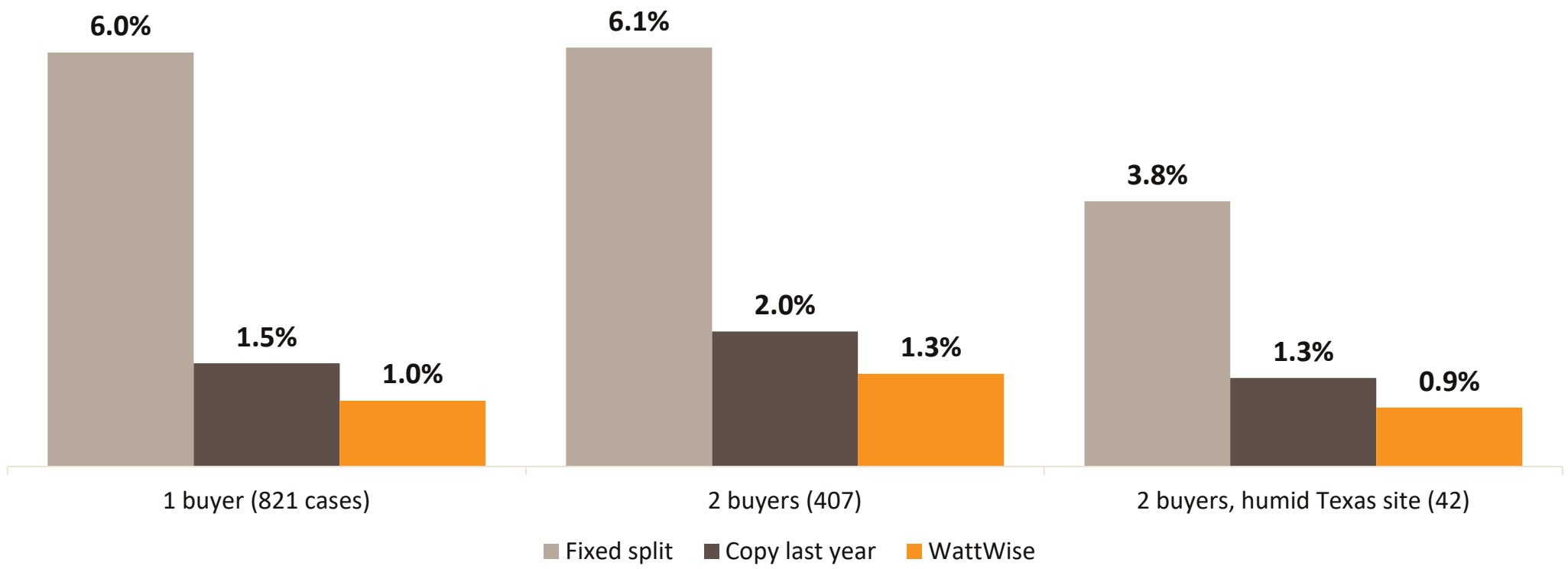
FROM TEAM PROMAX,

THANK YOU FOR YOUR TIME!

Completed: 23 Oct 2026

APPENDIX A: DOES IT REALLY REDUCE SOLAR GIVEAWAY?

Median share of the year's solar given away



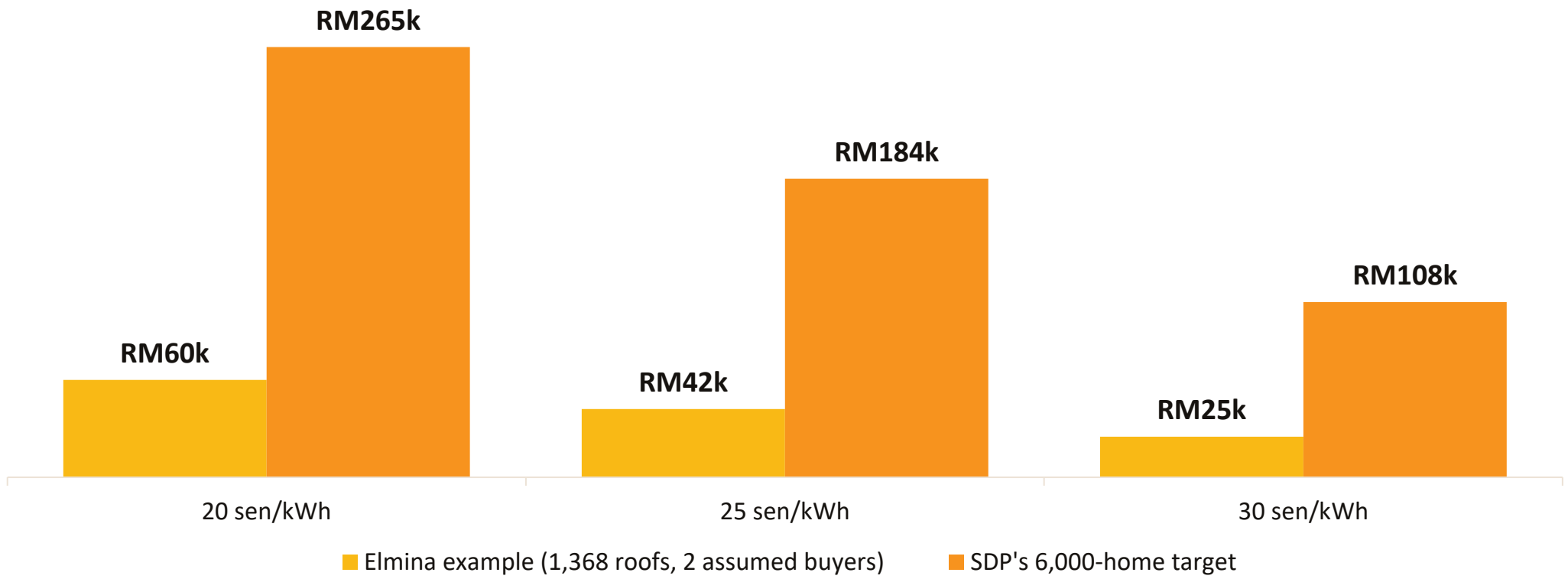
Fixed split 6.05% → WattWise 1.34% for buyer pairs.

Test it yourself

Real meters, 821 buildings (Miller et al., 2020); rooftop solar at Elmina (European Commission JRC, 2022); WattWise (2026).

APPENDIX B: HOW MUCH INCOME DOES IT ADD FOR SDP?

Extra gross scheme income a year, by assumed CREAM price



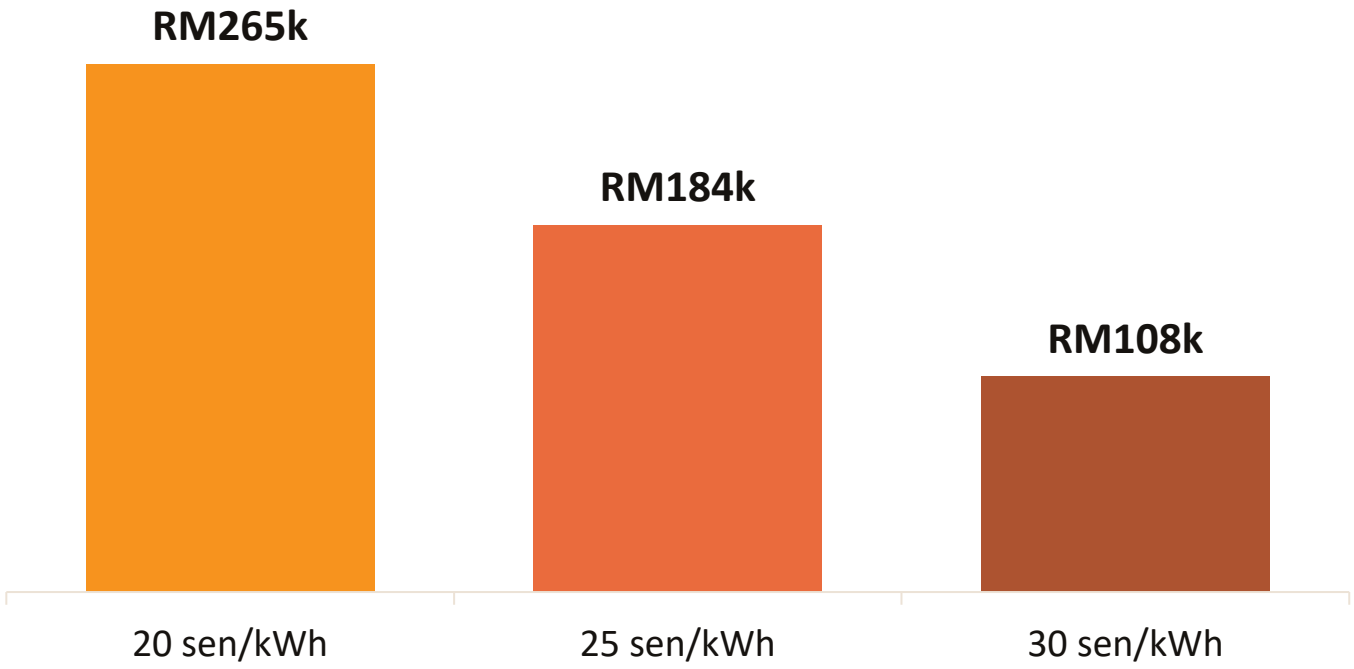
Modest and price-dependent; roof-lease terms are SDP's private deal.

Test it yourself

Modelled, 2 buyers assumed. Before roof-lease payments, costs and tax. At 15 sen the gain would be larger, but 15 sen is below the 17 sen NEDA break-even.

APPENDIX B (CONTINUED): HOW MUCH AT SDP'S 6,000-HOME TARGET?

What WattWise adds to a 6,000-home scheme (modelled)



Extra gross income a year vs a fixed split, by assumed CREAM price

Scales with solar generated: about RM2.6k–6.4k per 1,000 MWh a year (30 to 20 sen). More roofs mean more income; more buyers do not.

RM108k–265k

a year more than a fixed yearly split

1.4%–6.6%

of the scheme's total yearly income (RM4.0M–7.9M)

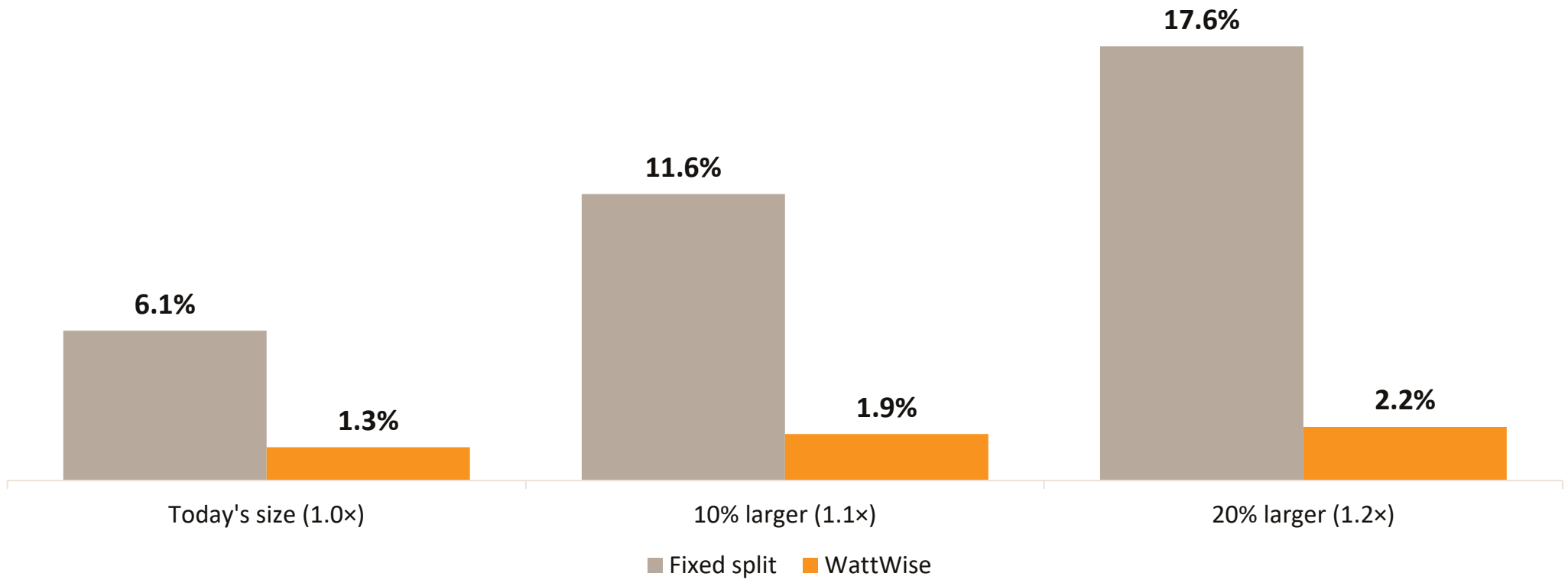
6,000 homes (≈41,000 MWh a year), 2 buyers assumed. Before roof-lease payments, costs and tax.

Modelled; the CREAM price is private (assumed 20–30 sen/kWh). Scheme income is after the 9 sen grid access charge. Below 17 sen, selling to NEDA beats selling to a buyer.

Check these numbers: Appendix B (extra income by CREAM price and scheme size).

APPENDIX C: WILL THERE REALLY BE SPACE FOR 20% MORE CREAM CONTRIBUTORS?

Solar given away when the solar pool is built larger



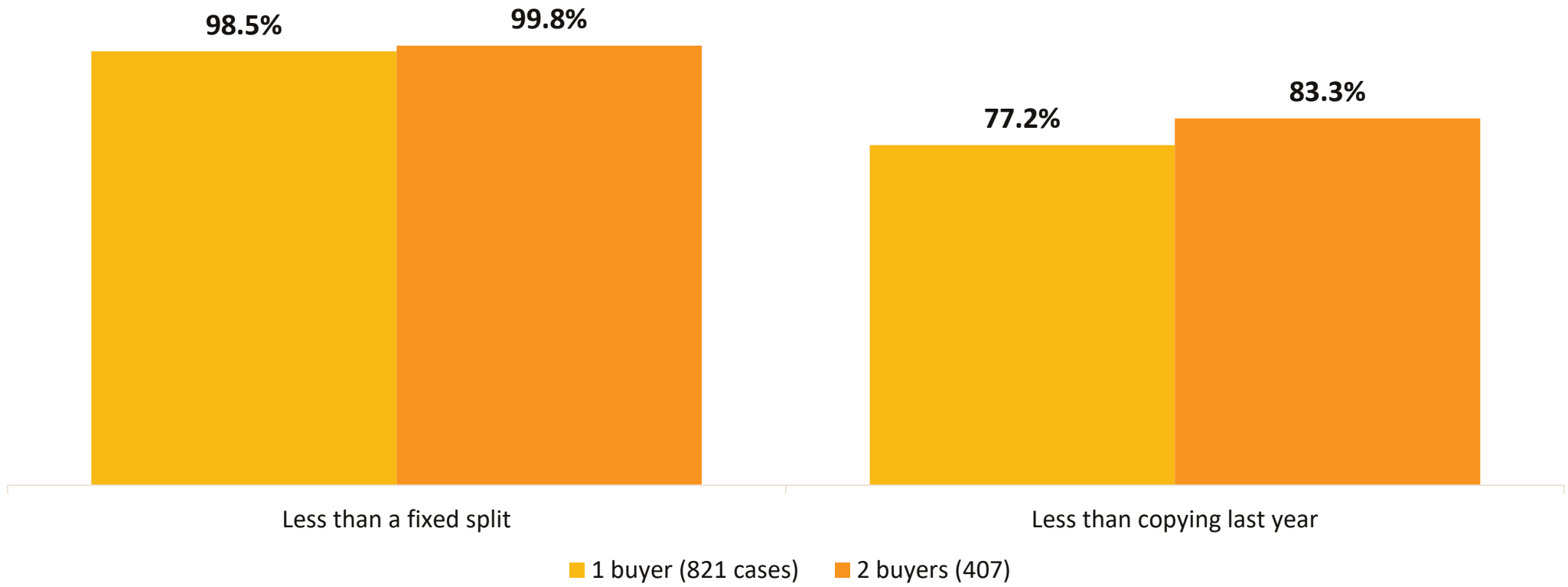
Still below today's fixed split, but ~76% of the extra solar sells via NEDA at 8 sen.

Test it yourself

Feeder limits (2 MWp each), roof availability and lease take-up are not modelled (WattWise, 2026).

APPENDIX D: DYNAMIC SPLITTING WORKS BETTER THAN FIXED ONES?

Share of cases where WattWise gives away less



Beats fixed in almost every case; above ~20 sen, copying last year earns slightly more.

Test it yourself

Real meters, 821 buildings (Miller et al., 2020); WattWise (2026).

APPENDIX E: REFERENCES

Energy Commission (2025) Guidelines for Community Renewable Energy Aggregation Mechanism (CREAM), GP/ST/No.55/2025. Putrajaya: Suruhanjaya Tenaga.

Energy Commission (2026) Grid Emission Factor (GEF) in Malaysia, 2022–2024 (Provisional). Putrajaya: Suruhanjaya Tenaga, 23 February. https://myenergystats.st.gov.my/documents/d/guest/grid-emission-factor-gef-in-malaysia-2022-2024-provisional-

European Commission Joint Research Centre (2022) Photovoltaic Geographical Information System (PVGIS) 5.2. https://re.jrc.ec.europa.eu/pvg_tools/

Miller, C. et al. (2020) 'The Building Data Genome Project 2', Scientific Data, 7, 368. https://doi.org/10.1038/s41597-020-00712-x

Selan, S. (2026) ‘Malaysia’s power grid strains under data centre boom and renewables growth’, Eco-Business, 15 September. https://www.eco-business.com/news/malaysias-power-grid-strains-under-data-centre-boom-and-renewables-growth/

Sime Darby Property (2026a) Integrated Annual Report 2025. Sime Darby Property (2026b) Sustainability Report 2025.

Skrine (2025) Energy Commission issues guidelines for Malaysia's CREAM [Legal alert]. April.

The Malaysian Reserve (2025) 'Sime Darby Property, TNB launch pilot solar project under CREAM to power homes, industries', 26 September.

WattWise (2026) WattWise code and results. https://github.com/ttc-099/Wattwise. Every figure is rebuilt by python replay.py in 02 Scheduler Module.

References in the testing suite